

STUDENT COMPETITION 2026
Sovereign Minigrid Electrification & Smart Street-EMS Challenge

Moving beyond standard cloud-centric hackathons, this competition is anchored around the open-source **Street-EMS** energy management platform. Grounded in the spirit of IEEE Smart Village technology workshops, participants tackle real-world rural minigrid architectures combining physical power engineering with business-case and shipping sensitivity.

Street-EMS
OPEN CORE FRAMEWORK

THE WORKSHOP TRACK

"MAKE, CODE, INTEGRATE" FORMAT

Delivered across three mentoring-driven sessions during the conference week, this track prepares in-country engineers and student teams to design, assemble, and program resilient minigrid interfaces.

- 1

MAKE — Build the Physical EMS Controller
Assemble a fully functional Street-EMS controller from PCB components. Learn assembly, testing, and hardware debug processes.
- 2

INTEGRATE — Cabinet Wiring & Multi-Tenant Metering
Mount the controller inside a 3-phase Street-EMS reference cabinet with DIN rail breakers, contactors, and multi-tenant metering variants.
- 3

CODE — Cloud vs. Cloudless Aggregation Challenge
Network Street-EMS under alternative cloud and cloudless protocols including DLMS/COSEM, the EnAccess REST API, and OpenEMS frameworks.

ELIGIBLE CANDIDATES

AFRICAN INSTITUTIONS

All competing authors must be registered students (Undergraduate or Postgraduate) attending an institution of learning in Africa at the time of submission. Submissions allow individual or group teams of up to 3 members.

IMPORTANT DATES

DEADLINES & TIMELINES

Call for Project Submissions	20th March 2026
Due Date for Submissions	4th July 2026 EXTENDED
Online Technical Interviews	15th July 2026
Final Selection & Notifications	20th July 2026
Award Ceremony & Live Final	During PAC 2026

CASH PRIZES & PLAQUE AWARDS

TOP CATEGORIES

POSTGRADUATE (PG)

1st:	\$500 + Plaque
2nd:	\$300 + Plaque
3rd:	\$200 + Plaque

UNDERGRADUATE (UG)

1st:	\$300 + Plaque
2nd:	\$200 + Plaque
3rd:	\$100 + Plaque

 **Travel Grants:** Top 1st place winners receive a fully covered travel, accommodation, and food grant to Nairobi, Kenya!

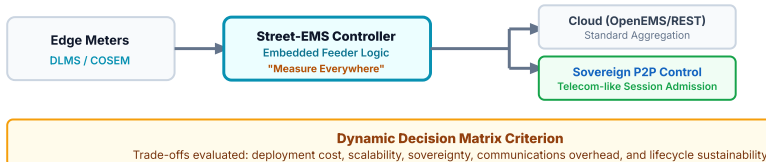
CORE ARCHITECTURE CHALLENGE

FIRST-KILOMETER GRID DISTRIBUTED INTELLIGENCE

We introduce a **truly cloudless, sovereign village electrification architecture** option based on Street-EMS embedded LV Feeder distributed intelligence.

Shifting the LV feeder from the traditional "last kilometer" to the **"first kilometer"** of community-owned, villager-empowered resilient systems.

Key Design Criteria: Measure everywhere, policy-based session admission, telecom-like P2P asset transfer control planes, and transmit-only actionable telemetry.



IN-COUNTRY STARTUPS TRACK & FREE TRADE LOGISTICS

Startups and engineers will gain critical insights into leveraging recent cross-border **African Free Trade Agreements (AfCFTA)**. We introduce proper usage of the **Harmonized System (HS) shipping codes** for modules and PCB import, assembly, test, programming, repair, and export—enabling commercial viability.

HS Code 85.37
ELECTRONIC CONTROLLERS

ATTENDEE TAKEAWAYS (DONOR FUNDED)

Take-Home EMS PCB

The first **15–20 registered attendees** receive a fully functional, assembled, and programmed Street-EMS PCB kit to take home.

Cabinet Donation

3–5 reference cabinets (single/three phase multi-tenant under 25 lbs) built during the track are donated to in-country institutions.

SUBMIT YOUR PROPOSAL

Register online and submit your team's structural design proposal. Teams selected for finals receive travel grants.

ieee-powerafrica.org/isvx-student-paper-contest/

Apply: forms.gle/DEtAb88AxCbweRH7

